

RETAIL BUSINESS ANALYTICS

Using SQL, Python & Power BI

A Business Intelligence Consulting Report

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Date: _____

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Executive Summary

Retail organizations today generate vast volumes of transactional data every day — sales, inventory movement, customer interactions, and product returns — yet many still make critical decisions based on intuition rather than evidence. In an industry defined by thin margins, seasonal volatility, and intense competition, the ability to convert raw operational data into timely, actionable insight has become a genuine source of competitive advantage.

Businesses that adopt a data-driven operating model consistently outperform peers on the metrics that matter most to leadership: revenue growth, profitability, inventory efficiency, and customer retention. Data-driven decision-making replaces guesswork with evidence, enabling management to identify where value is being created, where it is being eroded, and where the greatest opportunities for improvement lie.

This engagement was undertaken to build an end-to-end retail analytics capability for Maven Market, a multi-region grocery retail operation spanning the United States, Mexico, and Canada. The project integrates three complementary technologies into a single analytical workflow: Microsoft SQL Server for structured data consolidation and business querying, Python for deeper statistical analysis and exploratory visualization, and Power BI for interactive executive dashboards. Together, these tools transform disconnected transactional records into a coherent, decision-ready view of the business.

The result is a analytics foundation that allows leadership to move from reactive reporting to proactive management — monitoring performance in real time, diagnosing the drivers behind revenue and profit trends, and prioritizing initiatives with the greatest business impact.

This report is structured to support a range of executive decisions, including:

- Where to focus commercial investment across regions, stores, and product categories
- Which customer segments and products generate disproportionate value
- How operational issues such as returns are affecting profitability
- What strategic actions will most efficiently improve business performance

The remainder of this document sets out the business context, the analytical approach taken, and — in later sections — the detailed findings, dashboard-based insights, and executive recommendations arising from the analysis.

Business Problem

Retail businesses operating across multiple regions and store formats face a recurring set of structural challenges that make consistent, profitable growth difficult to sustain. Left unaddressed, these challenges compound over time and erode both revenue and margin.

Common Industry Challenges

- **Declining Profitability:** Rising cost of goods, promotional discounting, and operational overheads steadily compress margins even when top-line revenue appears stable.
- **Inefficient Inventory & Product Allocation:** Without visibility into which products actually drive revenue and profit, businesses over-invest in slow-moving stock and under-invest in high performers.
- **Regional Sales Disparities:** Performance frequently varies widely by geography, store type, and local market conditions, yet these differences are often invisible in aggregated reporting.
- **Unclear Customer Purchasing Behavior:** Many retailers cannot distinguish high-value, loyal customers from occasional shoppers, limiting the effectiveness of marketing and retention spend.
- **Inconsistent Product Performance Visibility:** Leadership often lacks a reliable, unified view of which products, brands, and categories are genuinely profitable versus merely high-volume.
- **Uncontrolled Returns:** Product returns quietly erode profitability and signal underlying quality, supply chain, or customer-expectation issues that go unaddressed without dedicated tracking.

Individually, each of these issues is manageable. Collectively, and without an integrated analytical view, they create a business environment where decisions are made on incomplete information — and where problems are only identified after they have already affected financial results.

How This Project Addresses These Challenges

This project was designed specifically to close that gap. By consolidating transactional, product, customer, store, and returns data into a single analytical environment, and by layering SQL-based querying, Python-based statistical analysis, and Power BI dashboards on top of that foundation, the project gives management a unified, evidence-based view of business performance — one that supports faster, more confident decision-making across every function from merchandising to store operations to customer strategy.

Business Objectives

The project was scoped around a defined set of business objectives, each directly tied to a decision that leadership needs to make regularly. These objectives guided every stage of the analytical workflow, from data preparation through to dashboard design.

- **Measure Overall Business Performance:** Establish a reliable, single source of truth for core performance indicators — revenue, profit, order volume, and margin — so leadership can track the health of the business at a glance.
- **Identify Revenue Drivers:** Determine which products, brands, stores, and regions contribute most to top-line revenue, enabling more targeted commercial focus.
- **Understand Customer Behavior:** Analyze purchasing patterns, order frequency, and customer value to inform marketing, loyalty, and retention strategy.
- **Evaluate Product Performance:** Assess profitability and sales concentration across the product catalog to guide assortment and inventory decisions.
- **Compare Regional Sales:** Surface performance differences across countries, regions, and store locations to identify expansion opportunities and underperforming markets.
- **Monitor Profitability:** Track profit and margin trends over time and across business dimensions, rather than relying on revenue alone as a success measure.
- **Analyze Returns:** Quantify the financial and operational impact of product returns and identify the products and stores most affected.
- **Build Executive Dashboards:** Translate analytical findings into intuitive, interactive Power BI dashboards that management can use without technical assistance.
- **Generate Actionable Recommendations:** Convert every analytical finding into a specific, business-relevant recommendation rather than leaving insights as abstract observations.

Dataset Overview

The analysis draws on a structured retail dataset covering two full fiscal years (1997–1998) of transactional activity across Maven Market's store network. Seven core datasets were combined to build a complete, analysis-ready view of the business.

Transactions

Purpose: The transactional backbone of the project, capturing every individual sales line across the two-year period — approximately 269,720 records in total. **Important fields** include transaction date, product, customer, store, and quantity sold. **Business importance:** This dataset is the foundation for every revenue, profit, and trend calculation in the report.

Products

Purpose: Describes the full product catalog — roughly 1,560 unique products across 111 brands. **Important fields** include product name, brand, cost, and retail price. **Business importance:** Enables profitability, margin, and category-level analysis, and supports product mix and inventory decisions.

Customers

Purpose: Contains demographic and account information for over 10,280 registered customers. **Important fields** include location, marital status, occupation, and membership tier. **Business importance:** Powers customer segmentation, loyalty analysis, and geographic demand mapping.

Stores

Purpose: Details the 24 physical store locations operated across the network. **Important fields** include store type, size, city, and country. **Business importance:** Enables store-level performance benchmarking and format-level comparison (e.g., Supermarket vs. Deluxe Supermarket).

Regions

Purpose: Maps stores to their broader sales regions and geographic markets. **Business importance:** Supports regional performance comparison and informs market expansion or consolidation decisions.

Calendar

Purpose: A supporting date dimension used to align transactions to fiscal periods, quarters, and months. **Business importance:** Underpins all time-based trend and seasonality analysis.

Returns

Purpose: Captures approximately 7,087 return records across the two-year period. **Important fields** include product, store, and returned quantity. **Business importance:** Provides visibility into product quality issues and their direct impact on profitability.

Data Preparation & Methodology

Rather than analyzing each data source in isolation, the project followed a structured methodology to build a single, trustworthy foundation for analysis before any business questions were addressed.

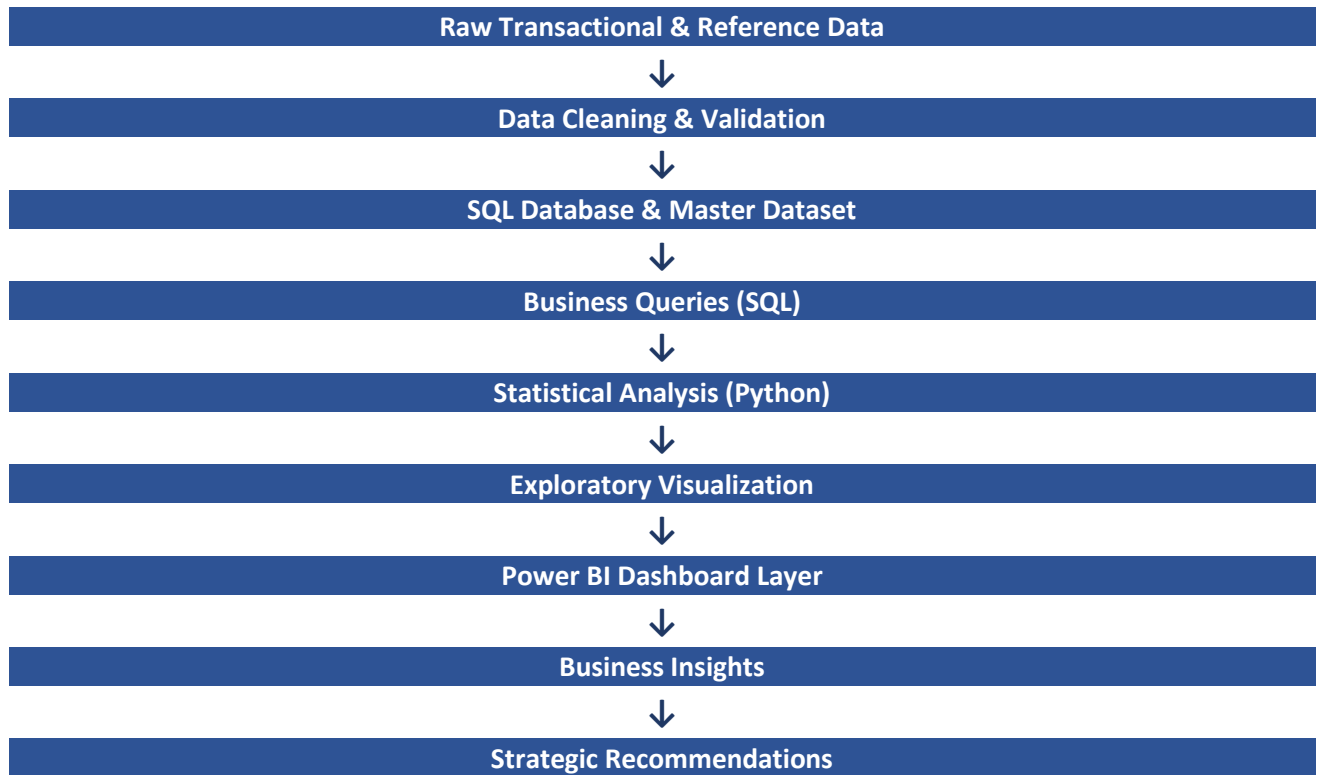
Overview of the Analytical Process

- **Data Collection:** Raw operational extracts covering transactions, products, customers, stores, regions, calendar, and returns were gathered from source systems.
- **Combining Yearly Transaction Files:** 1997 and 1998 transaction records were unified into a single continuous transaction history to enable accurate year-over-year comparison.
- **Data Cleaning:** Records were reviewed for consistency, and fields were standardized to ensure reliable joining and aggregation across datasets.
- **Data Validation:** Row counts, relationships, and key business totals were cross-checked across SQL and Python outputs to confirm the integrity of the combined dataset.
- **Table Joins:** Transaction records were linked to product, customer, store, and regional information to create a complete, enriched view of every sale.
- **Feature Creation:** New business-relevant fields — including sales value and profit per line item — were derived to support downstream analysis.
- **Sales & Profit Calculation:** Revenue was calculated from quantity and retail price; profit was calculated as the margin between retail price and product cost, multiplied by quantity sold.
- **Master Dataset Creation:** All sources were merged into a single master analytical table spanning 269,720 transaction records and 49 combined attributes, forming the foundation for both SQL and Python analysis.
- **SQL Querying:** Structured business queries were used to calculate core KPIs, rank performance across products, stores, customers, and regions, and surface trends using window functions.
- **Python Analysis:** Statistical and visual analysis in Python was used to validate SQL findings, explore distributions, and identify patterns such as revenue concentration and regional profitability relationships.
- **Dashboard Creation:** Findings were translated into interactive Power BI dashboards covering sales, orders, profit, returns, customers, and store performance for ongoing executive use.

This staged approach ensured that every downstream insight — from a single KPI to a strategic recommendation — is traceable back to a validated, well-understood data foundation.

Analytics Workflow

The project follows a linear, auditable workflow that carries data from raw operational extracts through to executive decision-making. Each stage adds a layer of business meaning to the underlying data.



This workflow ensures consistency: every number presented on a Power BI dashboard can be traced back through the Python analysis to the underlying SQL queries and, ultimately, to the source transactional data.

Report Roadmap

This document is being delivered in stages to allow for structured review at each milestone. Having established the business context, objectives, and analytical foundation in this section, the remainder of the report will build toward a complete set of findings and recommendations.

The sections that follow will cover:

- **SQL Analysis:** Core business KPIs and performance queries covering sales, products, customers, stores, and returns.
- **Python Analysis:** Deeper statistical exploration, trend analysis, and business analytics techniques such as Pareto and margin analysis.
- **Power BI Dashboards:** Interactive executive dashboards that bring the analysis to life for day-to-day business use.
- **Business Insights:** A consolidated view of what the data reveals about the business, organized by theme.
- **Strategic Recommendations:** Specific, actionable recommendations for leadership, each linked to a supporting observation and expected business impact.

Together, these sections will complete the transition from raw data to a fully actionable business intelligence report.

Chapter 5: Business Performance Analysis

SQL and Python were used together to evaluate Maven Market's performance across every major dimension of the business — from top-line revenue down to store-level profitability. SQL provided the structured querying layer needed to calculate consistent, auditable KPIs, while Python was used to validate these figures and explore them through statistical and visual analysis.

Key performance indicators (KPIs) give leadership a fast, reliable way to gauge the overall health of the business without needing to review granular transactional data. When tracked consistently over time, KPIs make it possible to spot emerging problems early, measure the impact of strategic initiatives, and hold different parts of the business to a common performance standard.

Core Business KPIs

\$1.76M Total Revenue (1997–98)	\$1.05M Total Profit (1997–98)	59.7% Overall Profit Margin
269,720 Total Transactions	833K Total Products Sold	\$6.54 Average Order Value
10,281 Registered Customers	24 Total Stores	111 Total Brands

What Each KPI Measures and Why It Matters

- **Total Revenue:** Measures the total value of goods sold. It is the primary indicator of commercial scale and top-line growth, and the figure most closely watched by executive leadership and investors.
- **Total Profit & Profit Margin:** Measures how much of that revenue converts into bottom-line value after cost of goods. Margin is a more reliable indicator of business health than revenue alone, since revenue growth without margin discipline can mask deteriorating unit economics.
- **Total Transactions & Products Sold:** Measures operational scale and customer engagement. Sustained growth here signals a healthy, active customer base rather than one relying on a small number of large orders.
- **Average Order Value (AOV):** Indicates typical basket size. Management uses AOV to evaluate the effectiveness of upsell, cross-sell, and pricing strategies.
- **Customer, Store, and Brand Counts:** Provide the denominators against which performance is normalized — allowing like-for-like comparison of revenue per customer, per store, and per brand.

Taken together, these KPIs confirm that Maven Market operates a high-volume, low-basket-value grocery model — profitability is driven by consistent, repeat transaction volume across a broad customer base rather

than by a small number of high-value purchases. This has direct implications for how management should prioritize growth initiatives, discussed further in the sections that follow.

Chapter 6: Sales Performance Analysis

Sales performance was evaluated over the full 1997–1998 period using both SQL trend queries and Python time-series visualization. The analysis focused on identifying the overall trajectory of the business, seasonal patterns, and the periods that most influenced full-year results.

Overall Sales Performance

Total revenue more than tripled between 1997 and 1998, growing from approximately \$565,000 in 1997 to approximately \$1.2 million in 1998. This represents a substantial acceleration in commercial performance rather than steady, incremental growth.

Monthly and Seasonal Trends

Monthly revenue in 1997 was relatively flat and modest, fluctuating in a narrow band between roughly \$42,000 and \$50,000 per month. Beginning in November and December 1997, revenue rose sharply, and this elevated level was sustained and further built upon throughout 1998, with monthly revenue settling into a consistently higher range of approximately \$95,000 to \$120,000.

High-Performing and Low-Performing Periods

- **High-Performing Periods:** The final quarter of 1998 was the strongest period in the dataset, with quarterly sales reaching approximately \$326,000 — the highest of any quarter analyzed. Late 1997 also marked a clear inflection point where the business stepped up to a materially higher sales run-rate.
- **Low-Performing Periods:** The first three quarters of 1997 were consistently the weakest, with quarterly revenue in the \$130,000–\$140,000 range — roughly half the level achieved in equivalent 1998 quarters.

Revenue Growth Patterns

The growth pattern is a step-change rather than a gradual ramp: performance shifts to a distinctly higher plateau at the 1997–1998 boundary and continues to build modestly from there, with Quarter 4 consistently the strongest quarter in both years. This points to a structural change in the business — such as store network expansion, an assortment change, or a demand shift — rather than organic, incremental growth alone.

Business Implications

The scale of the year-over-year improvement suggests the business found meaningful commercial traction that should be understood and, where possible, replicated. At the same time, the consistent Q4 strength across both years indicates a seasonal pattern that should inform staffing, inventory, and promotional planning.

Managerial Insights

- Treat late 1997 as a case study: identify what changed operationally or commercially at that inflection point and assess whether it can be deliberately reproduced in other markets.
- Plan inventory and staffing around a reliable Q4 peak, rather than a flat annual assumption, to avoid stock-outs during the strongest trading period.

- Investigate why Q1–Q3 1997 underperformed relative to the equivalent 1998 quarters, to rule out one-off issues versus a genuine baseline improvement.
- Use the 1998 monthly run-rate, not the 1997 average, as the basis for forward revenue planning and store-level targets.
- Monitor early-quarter performance closely each year, since the data shows this is where the business is most vulnerable to underperformance.

Chapter 7: Product Performance Analysis

Product-level performance was assessed using both SQL revenue and margin queries and Python-based Pareto analysis, drawing on the full catalog of roughly 1,560 products across 111 brands.

Best-Performing Products

Revenue at the individual product level is notably well distributed rather than concentrated in a small number of hero items. The top-performing products — including Hermanos Green Pepper, Hilltop Mint Mouthwash, Carlson Head Cheese, Great Pumppernickel Bread, and Fabulous Strawberry Drink — each contribute a broadly similar share of total revenue, in the region of 0.13%–0.14% per product, rather than a handful of products dominating sales.

Revenue Contribution and Product Concentration

The Pareto (80/20) analysis conducted in Python confirms this pattern: cumulative revenue builds gradually and near-linearly across the ranked product list, rather than showing the sharp early concentration typical of a classic 80/20 distribution. In practical terms, reaching 80% of total revenue requires a substantially larger share of the catalog than the traditional Pareto rule of thumb would suggest, indicating that Maven Market's revenue base is broad rather than narrow.

Product Profitability

Profit margins across the top-revenue products are consistently strong, generally ranging from roughly 51% to 67%, with products such as Hermanos Green Pepper and Super Extra Chunky Peanut Butter achieving margins above 67%. Importantly, high-revenue products are not systematically low-margin — the analysis did not surface a meaningful population of high-volume, low-profitability products that would warrant repricing or delisting on margin grounds alone.

Inventory and Product Mix Implications

Because revenue and profitability are both broadly distributed across the catalog, inventory risk is naturally diversified: no single product's underperformance would materially damage overall results. However, this also means that assortment-based growth strategies must work across many products simultaneously rather than relying on a small set of flagship items.

Managerial Insights

- Do not over-index merchandising or promotional investment on a small set of 'hero' products — the data shows performance is broad-based, and disproportionate investment in a few items may not reflect underlying demand patterns.
- Use the broad margin strength (51%–67% across top products) as confirmation that pricing discipline is generally sound; focus margin-improvement efforts on the lower-revenue, lower-margin tail rather than the top performers.
- Given the flat revenue distribution, prioritize category-level (rather than single-SKU) strategies for assortment planning and supplier negotiation.

- Periodically re-run the Pareto analysis to detect whether concentration increases over time — an early indicator of shifting customer preferences.

Chapter 8: Customer Analysis

Customer-level analysis combined SQL ranking queries with Python-based segmentation to understand how the customer base contributes to overall business performance.

Customer Purchasing Behavior

With over 10,280 registered customers and roughly 8,840 of them transacting during the analysis period, Maven Market operates a broad, high-frequency customer base rather than one reliant on a small number of large accounts. The average order value of \$6.54 confirms a grocery-style purchasing pattern built on frequent, modest-value visits rather than infrequent, large-basket purchases.

High-Value Customers

Even the highest-spending individual customers — topping out at roughly \$4,470 in total revenue over the two-year period — represent a very small fraction of total company revenue. No single customer or small customer group exerts outsized influence on overall results, which reduces revenue concentration risk but also means that traditional 'VIP customer' retention programs will have limited standalone impact on total revenue.

Customer Segmentation

A tiered segmentation model (Bronze, Silver, and Gold, based on cumulative revenue thresholds) was applied to the customer base. The overwhelming majority of customers fall into the Bronze tier, reflecting modest average spend per customer — consistent with a grocery retail model where value is built through purchase frequency and basket consistency rather than large individual transactions.

Customer Contribution and Purchase Frequency

Revenue and profit contribution by customer track closely together, indicating consistent margin behavior across the customer base rather than a subset of customers being served at disproportionately low profitability. Geographic analysis further shows customer revenue concentrated in Washington, California, and Oregon, mirroring the store network's North American footprint.

Business Importance of Understanding Customer Behavior

Because value is distributed across a large, frequent-purchasing customer base rather than a small number of high-value accounts, customer strategy should prioritize broad-based retention and frequency-building initiatives over narrow VIP-focused programs. Understanding this distinction directly shapes where marketing and loyalty budgets will generate the greatest return.

Managerial Insights

- Prioritize customer retention and visit-frequency programs over high-value account management, given the broad, low-concentration nature of customer revenue.
- Use membership card tiers (Bronze, Normal, Golden, Silver) as a foundation for structured loyalty incentives tied to purchase frequency rather than basket size.

- Focus geographic customer-acquisition investment on the states and regions already showing the strongest customer density — Washington, California, and Oregon — before expanding into lower-density markets.
- Track customer segment migration (Bronze to Silver to Gold) over time as a leading indicator of overall customer base health.

Chapter 9: Store & Regional Performance

Store- and region-level analysis was conducted using SQL aggregation queries and Python's regional performance matrix, covering all 24 stores across the United States, Mexico, and Canada.

Store Comparison

Performance across the 24-store network is uneven. Store 13 and Store 17 are the clear top performers, generating approximately \$170,000 and \$158,000 in revenue respectively — meaningfully ahead of the rest of the network. A second tier of stores (Store 15, 11, 24, 7, 3, and 16) cluster tightly together in the \$110,000–\$116,000 range, while the lowest-performing stores generate a small fraction of this — as little as roughly \$9,800 at the weakest location.

Regional Comparison

The North West region is by a wide margin the dominant contributor to company performance, generating roughly \$848,000 in revenue and \$506,000 in profit — nearly half of total company revenue from a single region. Mexico Central and South West form a clear second tier (each in the \$320,000–\$330,000 revenue range), while Canada West, Mexico South, Mexico West, and particularly Central West contribute comparatively little.

Sales Distribution and Operational Performance

At the country level, the United States accounts for the large majority of revenue, consistent with its larger store count (13 of 24 stores), followed by Mexico (9 stores) and Canada (2 stores). Profit tracks revenue closely at both the store and regional level, indicating that underperforming locations are a volume problem rather than a margin problem — profitability percentage is broadly consistent across strong and weak performers alike.

Geographic Differences and Expansion Opportunities

The heavy concentration of performance in the North West region, combined with a consistent margin profile across all regions, suggests that geographic expansion or reinvestment is more likely to succeed by replicating the North West's operating conditions elsewhere than by pursuing entirely new markets.

Managerial Insights

- Conduct a root-cause study of Store 13 and Store 17's outperformance (location, format, local demographics, catalog mix) to identify replicable success factors for underperforming stores.
- Treat the lowest-revenue stores (such as those generating under \$20,000 over the period) as candidates for a formal performance review, covering location viability, format fit, and potential resizing or closure.
- Given consistent margin percentages across regions, prioritize volume-driving initiatives — local marketing, extended hours, assortment localization — over margin-focused interventions in underperforming markets.
- Use the North West region's outsized contribution as the primary case study when evaluating new store locations or market entry.

- Monitor the concentration of revenue in a single region as a business risk; consider deliberate initiatives to grow the contribution of secondary regions to reduce dependency.

Chapter 10: Returns & Profitability Analysis

Returns and profitability were analyzed together, since returns directly affect both realized revenue and customer satisfaction. The analysis draws on approximately 7,087 return records and the corresponding SQL and Python profitability calculations.

Returns Overview

Across the two-year period, the business recorded roughly 8,289 returned orders, totaling approximately \$7,040 in returned value and 16,578 returned units. While the direct dollar value of returns is modest relative to total revenue, return activity increased notably from 1997 into 1998 — rising from around 650–730 units per half-year in 1997 to over 1,300–1,590 units per half-year in 1998 — tracking the broader growth in sales volume over the same period.

Products Most Affected by Returns

Returns are led by products such as Hermanos Red Pepper, Hermanos Oranges, and Walrus Merlot Wine in absolute unit terms. More significant from a business-risk perspective, several products show very high return rates as a percentage of units sold — including one product line returning at over 67% of units sold — indicating a possible quality, freshness, or fulfillment issue specific to those items rather than a general returns problem.

Profit and Profit Margin

Overall profitability remains strong: total profit of approximately \$1.05 million against total revenue of approximately \$1.76 million represents an overall margin of roughly 59.7%. Profit grew in line with revenue between 1997 and 1998, and the ratio of cost to revenue has remained stable month to month (cost consistently around 40% of sales, profit around 60%), indicating disciplined and consistent pricing rather than margin erosion over time.

Financial Health and Operational Efficiency

The combination of stable, healthy margins and a modest but rising returns rate suggests that overall financial health is sound, but that returns — while currently a small drag on results — merit proactive monitoring given their upward trajectory alongside sales growth.

Managerial Insights

- Prioritize a root-cause investigation into the small number of products with return rates exceeding 60%, as these are disproportionately damaging to both profitability and customer experience relative to their sales volume.
- Monitor the returns growth rate relative to sales growth going forward; a returns rate rising faster than sales would signal an emerging quality or fulfillment issue rather than simple scale effects.
- Maintain the current pricing and cost discipline that has kept the cost-to-revenue ratio stable — this is a genuine strength and should be protected as the business scales.
- Incorporate store-level return rates (Store 17 and Store 13 show the highest return order counts) into store performance reviews alongside revenue and profit metrics.

- Use product-level return-percentage data, not just return-volume data, to prioritize quality investigations — high-percentage, low-volume products are early warning signs that pure volume rankings would miss.

Chapter 11: Summary of Business Insights

This section consolidates the strongest findings from the combined SQL and Python analysis into a single set of executive insights. Each insight integrates evidence from both analytical tracks and is structured to move directly from observation to interpretation to recommended action.

1. Revenue Growth

Observation: Total revenue grew from approximately \$565,000 in 1997 to approximately \$1.2 million in 1998, more than doubling year over year.

Business Interpretation: The business achieved a structural step-change in performance rather than gradual organic growth, most likely driven by an operational or commercial shift around late 1997.

Recommended Action: Investigate the specific drivers behind the late-1997 inflection point and evaluate whether they can be deliberately replicated in underperforming regions.

2. Seasonal Concentration

Observation: Q4 is consistently the strongest quarter in both 1997 and 1998, while Q1–Q2 are consistently the weakest.

Business Interpretation: Demand is seasonally concentrated, creating both a planning opportunity and an inventory risk around year-end.

Recommended Action: Align inventory build-up, staffing, and promotional calendars to the confirmed Q4 peak rather than a flat annual assumption.

3. Product Revenue Distribution

Observation: Revenue is spread broadly and evenly across the product catalog rather than concentrated in a small set of top sellers.

Business Interpretation: Growth strategies focused on a handful of 'hero' products are unlikely to move total revenue meaningfully; the opportunity is broad-based.

Recommended Action: Shift assortment and promotional strategy toward category-level initiatives rather than single-SKU campaigns.

4. Product Profitability

Observation: Top-revenue products carry healthy margins, generally between 51% and 67%.

Business Interpretation: High sales volume and strong margin are not in tension at Maven Market — there is no evidence of a 'sell more, earn less' trade-off among top products.

Recommended Action: Continue current pricing discipline on top-performing products; redirect margin-improvement effort toward lower-performing, lower-margin items instead.

5. Customer Base Structure

Observation: The vast majority of the 10,281 registered customers fall into the lowest ('Bronze') revenue segment, and even the top individual customers contribute a small share of total revenue.

Business Interpretation: Business value is generated through broad, frequent, low-basket-value transactions rather than a small number of high-value customer relationships.

Recommended Action: Prioritize frequency- and retention-focused loyalty programs over high-touch VIP account management.

6. Customer Geographic Concentration

Observation: Customer revenue is concentrated in Washington, California, and Oregon.

Business Interpretation: The customer base closely mirrors the store network's geographic footprint, suggesting store presence is a primary driver of customer acquisition.

Recommended Action: Use existing high-density markets as templates when evaluating future store or marketing investment locations.

7. Store Performance Variation

Observation: Store 13 and Store 17 significantly outperform the rest of the 24-store network, while the lowest-performing stores generate only a small fraction of top-store revenue.

Business Interpretation: A meaningful share of the network is materially underperforming relative to its peers, representing both a risk and an improvement opportunity.

Recommended Action: Conduct a structured performance review of the lowest-revenue stores, informed by a root-cause study of what makes Store 13 and Store 17 successful.

8. Regional Revenue Concentration

Observation: The North West region generates nearly half of total company revenue and profit, far ahead of any other region.

Business Interpretation: The business carries meaningful geographic concentration risk, with performance heavily dependent on a single region.

Recommended Action: Pursue deliberate growth initiatives in secondary regions (Mexico Central, South West) to diversify the revenue base over time.

9. Consistent Regional Margins

Observation: Profit margin percentage is broadly consistent across regions, even though absolute revenue varies widely.

Business Interpretation: Underperformance in weaker regions and stores is a volume issue, not a pricing or cost-control issue.

Recommended Action: Focus regional turnaround plans on demand generation and store operations rather than pricing or margin interventions.

10. Overall Profitability

Observation: The business achieved an overall profit margin of approximately 59.7%, with profit growing in step with revenue across the period.

Business Interpretation: Profitability is strong and stable, and the cost structure has not eroded as the business has scaled.

Recommended Action: Protect current pricing and cost discipline as a core strength while pursuing growth initiatives elsewhere in the business.

11. Returns Trend

Observation: Returned units roughly doubled from the first half of 1997 to the second half of 1998, broadly tracking sales growth.

Business Interpretation: Rising returns are, at this stage, a natural consequence of higher sales volume rather than a sign of deteriorating quality — but this should be actively monitored.

Recommended Action: Establish a recurring quarterly review comparing returns growth rate to sales growth rate to catch any future divergence early.

12. Return Rate Outliers

Observation: A small number of products show return rates in excess of 60% of units sold, far above the typical product.

Business Interpretation: These specific products likely have quality, freshness, or fulfillment issues that a pure revenue or volume view would not surface.

Recommended Action: Launch a targeted quality and supply-chain review of the highest return-percentage products, independent of their overall sales ranking.

13. Operational Efficiency Signal

Observation: Store-level profit tracks store-level revenue almost proportionally across the entire network.

Business Interpretation: Store operations are fundamentally sound across the business; the primary lever for improving weak stores is driving more customer volume, not fixing cost structure.

Recommended Action: Standardize store-level dashboards on volume and traffic-driving metrics as the primary lens for underperforming locations.

Chapter 12: Executive Dashboard Overview

While SQL and Python provided the analytical rigor behind this report, Power BI transforms those findings into a live, interactive decision-support tool for ongoing management use. Modern organizations increasingly rely on executive dashboards precisely because static reports go stale — a dashboard connected to the underlying data model allows leadership to monitor performance continuously rather than periodically.

The Maven Market Power BI dashboard is organized into six focused views — Sales Analysis, Orders Overview, Profit Analysis, Returns Analysis, Customers Insight, and Stores Performance — each built around the KPIs and business questions identified during the SQL and Python analysis phases. Interactive filtering (for example, by customer country) allows management to move fluidly from a company-wide view down to a specific market or segment without needing analyst support.

In this way, Power BI closes the loop on the analytical workflow: SQL established what the numbers are, Python explained why the patterns exist, and Power BI now puts both in front of decision-makers in a form built for daily and weekly business review rather than one-time analysis.

Chapter 13: Dashboard-Based Business Insights

Each dashboard view was designed around a specific business question, translating the analytical findings from earlier chapters into a format leadership can monitor continuously.

Sales Dashboard

Business Objective: Enable continuous monitoring of revenue performance, trend direction, and geographic sales distribution.

Key Metrics: Total sales by year and quarter, sales by customer country, sales by product brand, top-performing cities.

Business Value: Gives leadership an always-current view of revenue trajectory, removing the lag associated with periodic manual reporting and allowing early detection of slowdowns or acceleration.

Management Decisions Supported: Quarterly forecasting, promotional timing, and country-level commercial prioritization.

Customer Dashboard

Business Objective: Provide visibility into who Maven Market's customers are and how they engage with the business.

Key Metrics: Total customers (10.28K), customer country and state distribution, occupation mix, homeownership, gender and marital status, membership tier distribution.

Business Value: Translates the customer segmentation analysis into an operational tool for marketing and loyalty teams to target campaigns by demographic and geographic profile.

Management Decisions Supported: Loyalty program design, targeted marketing investment, and new-market customer profiling.

Product Dashboard

Business Objective: Surface which products and brands are performing best in terms of both orders and profitability.

Key Metrics: Top 5 most-sold products, top 10 brands by order volume, top 10 most profitable products, average sales value.

Business Value: Gives merchandising teams a fast way to identify which products merit expanded shelf space or promotional support, and which brands are driving order volume versus pure revenue.

Management Decisions Supported: Assortment planning, brand negotiation priorities, and promotional product selection.

Store & Regional Dashboard

Business Objective: Compare performance across the store network and geographic footprint to guide operational and expansion decisions.

Key Metrics: Top 10 stores by sales, store count and type by country, store establishment timeline, regional sales distribution.

Business Value: Makes store-level and regional disparities immediately visible to operations leadership, supporting faster intervention in underperforming locations.

Management Decisions Supported: Store performance reviews, expansion or consolidation planning, and regional resource allocation.

Returns & Profitability Dashboard

Business Objective: Monitor the financial and operational impact of returns alongside overall profit performance.

Key Metrics: Total orders returned (8,289), total returns value (\$7.04K), top returned products, top stores by returns, quarterly profit trend, cost-to-profit ratio by month.

Business Value: Connects returns activity directly to profitability tracking, allowing leadership to distinguish normal returns growth (tracking sales volume) from emerging quality issues.

Management Decisions Supported: Quality and supply-chain escalation, store-level accountability for return rates, and margin protection.

Chapter 14: Executive Recommendations

The following recommendations synthesize findings from the SQL analysis, Python analysis, and Power BI dashboards into a single set of practical actions for leadership, organized by strategic category.

Revenue Growth

Current Observation: Revenue nearly tripled from 1997 to 1998, with Q4 consistently the strongest quarter in both years.

Business Recommendation: Formalize the drivers behind the 1997–1998 growth step-change and build a repeatable playbook; align annual planning around the confirmed Q4 seasonal peak.

Expected Business Impact: A structured, evidence-based growth plan replacing ad hoc quarterly targets, reducing the risk of under- or over-investing in any given period.

Customer Strategy

Current Observation: Customer value is broad-based and frequency-driven rather than concentrated in a small number of high-spending accounts.

Business Recommendation: Shift loyalty and marketing investment toward frequency- and retention-building programs (e.g., visit-based rewards) rather than high-touch VIP account management.

Expected Business Impact: Improved return on marketing spend by aligning tactics with actual customer value distribution.

Product Strategy

Current Observation: Revenue and profitability are both evenly distributed across a broad product catalog, with no dominant low-margin, high-volume products.

Business Recommendation: Pursue category-level rather than single-SKU promotional and merchandising strategies; protect pricing discipline on the currently healthy 51%–67% margin range.

Expected Business Impact: More efficient use of promotional budget and reduced risk of over-investing in a small set of products that will not move total revenue materially.

Inventory Management

Current Observation: Sales are seasonally concentrated in Q4, and returns have grown alongside sales volume.

Business Recommendation: Build Q4-weighted inventory planning cycles and introduce return-rate monitoring as a standard inventory KPI, not just a post-sale metric.

Expected Business Impact: Reduced stock-out risk during peak season and earlier detection of products with disproportionate return activity.

Store Operations

Current Observation: Store 13 and Store 17 significantly outperform the rest of the 24-store network; several stores generate only a small fraction of top-store revenue.

Business Recommendation: Conduct a root-cause study of top-performing stores and apply findings through a formal improvement plan for the lowest-performing locations.

Expected Business Impact: Narrower performance gap across the store network and a clearer basis for future site-selection decisions.

Regional Expansion

Current Observation: The North West region generates nearly half of total company revenue, with performance heavily concentrated relative to other regions.

Business Recommendation: Pursue deliberate, measured growth initiatives in the second-tier regions (Mexico Central, South West) to reduce single-region dependency.

Expected Business Impact: A more diversified, resilient regional revenue base and reduced exposure to region-specific demand shocks.

Pricing Strategy

Current Observation: Profit margin percentage is stable both over time (around 59.7% overall) and across regions.

Business Recommendation: Maintain current pricing discipline as a core strength; avoid broad discounting strategies that would erode a margin profile that is already performing well.

Expected Business Impact: Continued margin protection as the business scales, avoiding the common pitfall of trading margin for volume unnecessarily.

Return Reduction

Current Observation: A small number of products show return rates exceeding 60% of units sold, well above the norm.

Business Recommendation: Launch a targeted quality and fulfillment investigation into the highest return-percentage products, independent of their sales ranking.

Expected Business Impact: Reduced returns-driven revenue and profit leakage, and improved customer satisfaction on affected product lines.

Operational Efficiency

Current Observation: Profit tracks revenue proportionally at both the store and regional level, indicating that underperformance is a volume issue rather than a cost-structure issue.

Business Recommendation: Focus operational improvement programs for weak stores and regions on demand generation (marketing, hours, localization) rather than cost-cutting.

Expected Business Impact: More effective allocation of operational improvement resources toward the interventions most likely to move performance.

Business Intelligence Adoption

Current Observation: The SQL, Python, and Power BI workflow built through this project currently exists as a project deliverable rather than an embedded business process.

Business Recommendation: Institutionalize the Power BI dashboard suite as a standing part of monthly and quarterly business reviews, with clear ownership for keeping the underlying data pipeline current.

Expected Business Impact: Sustained, ongoing value from the analytics investment rather than a one-time report that loses relevance as new data accumulates.

Chapter 15: Business Impact

Beyond the specific findings and recommendations detailed in this report, the analytical solution itself — the integrated SQL, Python, and Power BI workflow — represents a durable capability upgrade for Maven Market. Its value extends well beyond the initial analysis.

Revenue Management

With reliable, real-time visibility into revenue by product, store, and region, leadership can identify emerging trends — positive or negative — far earlier than would be possible through periodic manual reporting, enabling faster corrective or investment action.

Customer Decision Making

A clear, data-backed understanding of customer value distribution allows marketing and loyalty investment to be allocated to the strategies most likely to work, rather than relying on assumptions about which customers matter most.

Inventory Planning

Seasonal and product-level demand patterns, now clearly documented, provide a factual basis for inventory build-up and allocation decisions, reducing both stock-out and overstock risk.

Product Portfolio Management

Understanding that revenue and profitability are broadly distributed across the catalog — rather than concentrated in a few flagship products — changes how merchandising and category management should prioritize their attention going forward.

Regional Planning

Visibility into the concentration of performance in the North West region gives leadership an evidence base for decisions about where to defend market position and where to invest in growth or diversification.

Executive Reporting

The Power BI dashboard suite replaces fragmented, manually assembled reporting with a single, consistent source of truth — reducing the time executives spend reconciling numbers and increasing the time available for decision-making.

Data-Driven Decision Making

Perhaps most importantly, this project establishes a repeatable model for how Maven Market can approach future business questions: consolidate the data, validate it through structured querying and statistical analysis, and present it through an accessible dashboard — a pattern that can be extended to new business questions well beyond the scope of this initial engagement.

Chapter 16: Project Conclusion

This project set out to address a challenge common to multi-region retail businesses: the absence of a unified, evidence-based view of performance across sales, products, customers, stores, and returns. Without such a view, decisions are made on partial information, and problems are typically identified only after they have already affected financial results.

By integrating SQL for structured data consolidation and querying, Python for statistical validation and deeper exploratory analysis, and Power BI for accessible, interactive dashboards, this project delivered a complete analytical workflow — from raw transactional data through to executive-ready insight — built specifically around Maven Market's business questions.

The analysis surfaced a set of clear, consistent findings: a business with strong and stable profitability, a broad and resilient product and customer base, a pronounced seasonal (Q4) sales pattern, meaningful performance variation across stores, and a notable geographic concentration of revenue in the North West region. Each of these findings has been translated into specific, actionable recommendations for leadership across revenue growth, customer strategy, product management, store operations, and regional planning.

Business intelligence capability of this kind is no longer a discretionary investment for retail organizations — it is a foundational requirement for competing effectively in a market where margins are thin and customer expectations continue to rise. Retailers that build and sustain this capability are better equipped to respond quickly to changing conditions, allocate resources where they will have the greatest impact, and make the kind of confident, evidence-based decisions that compound into lasting competitive advantage.

This report, and the underlying analytical workflow behind it, is intended not as a one-time deliverable but as the foundation for an ongoing, data-driven approach to managing and growing the business.

Chapter 17: Future Scope

The analytical foundation established in this project can be extended in several directions to deepen its business value over time:

- **Sales Forecasting:** Building on the identified seasonal and growth patterns to generate forward-looking revenue projections.
- **Demand Prediction:** Applying predictive techniques at the product and store level to improve inventory and staffing precision.
- **Customer Lifetime Value Analysis:** Extending the current segmentation model to project long-term customer value and inform retention investment.
- **Market Basket Analysis:** Identifying product association patterns to inform cross-merchandising and promotional bundling.
- **Machine Learning Models:** Applying predictive modeling to areas such as churn risk, return likelihood, and price sensitivity.
- **Recommendation Systems:** Using purchase history to power personalized product recommendations across customer touchpoints.
- **Real-Time Dashboards:** Evolving the current Power BI suite from periodic refresh to real-time or near-real-time data connectivity.
- **Automated Data Pipelines:** Reducing manual effort in data consolidation through scheduled, automated ETL processes feeding the analytical environment.

Each of these extensions builds directly on the data foundation and analytical workflow established in this project, positioning Maven Market to evolve from descriptive and diagnostic analytics toward predictive and prescriptive capability over time.